

DAY 06 – IPv4 ADDRESSING FUNDAMENTALS

Networking Assignment Report

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Assignment Code	DAY06-IPV4-ADDRESSING
Total Marks	20
Operating System	Ubuntu Linux

Ubuntu Linux Note: The assignment specifies Windows commands such as `ipconfig` and `ipconfig /all`. This practical was performed on Ubuntu Linux, so the Linux equivalents `ip a` / `ifconfig -a` and `resolvectl status` were used in place of `ipconfig` and `ipconfig /all` respectively.

IPv4 ADDRESSING – THEORY

1. What is IPv4?

IPv4 provides logical addressing of the devices communicating over IP networks. The IPv4 address is 32 bits long, and it is common to represent it as four decimal numbers separated by dots, for instance: 192.168.1.10

2. What are octets in an IPv4 address?

The four numbers that comprise an IPv4 address (e.g. 192 . 168 . 1 . 10) are referred to as octets. There are four octets in an IPv4 address because $8 + 8 + 8 + 8 = 32$ bits in total.

3. What is the network portion and host portion of an IPv4 address?

An IPv4 address identifies both a network and a host on that network. The subnet mask specifies what portion of an IPv4 address belongs to the network, and what portion belongs to the host. Take, for instance, an IP address 192.168.1.10 and subnet mask 255.255.255.0 – in this case, 192.168.1 is the network portion, and 10 is the host portion.

4. What is a subnet mask?

A subnet mask specifies which portion of an IPv4 address refers to the network. One example of a subnet mask is 255.255.255.0 – another way to write it would be to use a prefix notation: 192.168.1.10/24 .

5. What are the three private IPv4 ranges?

The three private IPv4 ranges are:

10.0.0.0/8 (10.0.0.0 – 10.255.255.255)

172.16.0.0/12 (172.16.0.0 – 172.31.255.255)

192.168.0.0/16 (192.168.0.0 – 192.168.255.255)

These ranges are used in private networks, such as home, office, schools, or enterprise networks – they are not directly routable on the public Internet.

6. What is a public IPv4 address?

A public IPv4 address is a globally routable address. Internet-facing infrastructures use public IPv4 addresses – most of the time, a device in a private network will have a private IP address and a public IP address will be used by a modem/router in that same private network to access the Internet.

7. What is a loopback address?

The loopback range is 127.0.0.0/8 , with 127.0.0.1 (localhost) being the most common address in that range. It is used to bind a network connection to a local machine, which is useful when testing if the local TCP/IP stack is working correctly (e.g. with a command ping 127.0.0.1).

8. What is APIPA?

APIPA stands for Automatic Private IP Addressing. It is a range 169.254.0.0/16 – when a Windows machine cannot contact a DHCP server, it will assign itself an address from this range. Seeing a 169.254.x.x address on a machine is typically an indicator of a problem with the machine's network configuration.

9. What is a default gateway?

The default gateway is a router that connects a local network to other networks (such as the Internet). For instance, the default gateway for a PC with an IP address 192.168.1.10 could be 192.168.1.1 – this way, the PC can access the Internet.

10. What is the difference between static and dynamic IP addressing?

A static IP address is a manually assigned address that rarely changes. They are typically used for servers, printers, and other network equipment. A dynamic IP address is a temporarily assigned address, usually provided by a DHCP server.

11. What is the purpose of DHCP?

DHCP stands for Dynamic Host Configuration Protocol. It is used to dynamically assign IPv4 address, subnet mask, default gateway, and DNS server to a host.

12. What are network and broadcast addresses?

The network address refers to the network portion of an IPv4 address range. Take, for instance, an IPv4 address 192.168.1.10/24 – the network address for this range is 192.168.1.0. The broadcast address is the far end of an IPv4 address range, beyond the network address. For the example IPv4 address range given above, the broadcast address would be 192.168.1.255.

13. Why does IPv4 addressing matter in cybersecurity?

Cybersecurity analysts deal with source and destination IP addresses in log files of firewalls, servers, packet captures, intrusion detection and prevention systems, Security Information and Event Management (SIEM) platforms, Network Mapper (Nmap) scans, and Wireshark traffic captures among others. Having a good understanding of IPv4 addressing is a prerequisite to effectively tackle any of these areas.

PRACTICAL — EXPLORING IPv4 CONFIGURATION

The assignment requests ipconfig, ipconfig /all, ping 127.0.0.1, ping google.com, and nslookup google.com. Since this practical was performed on Ubuntu Linux, the equivalent commands ip a / ifconfig -a, resolvectl status, ping -c 4, and nslookup were used.

14. Recorded IPv4 configuration (Task 1)

Field	Value
Network adapter	wlp0s20f3 (Wi-Fi)
IPv4 address	172.20.10.3
Subnet mask	255.255.255.240
Default gateway	172.20.10.1
DNS server	172.20.10.1 (also fe80::e85f:2ff:fe2f:7664)
DHCP enabled	Yes (address auto-assigned, not statically configured)

15. Address classification (Task 2)

IP Address	Type
192.168.1.20	Private
10.10.10.5	Private
172.20.5.10	Private
127.0.0.1	Loopback
169.254.10.20	APIPA
8.8.8.8	Public

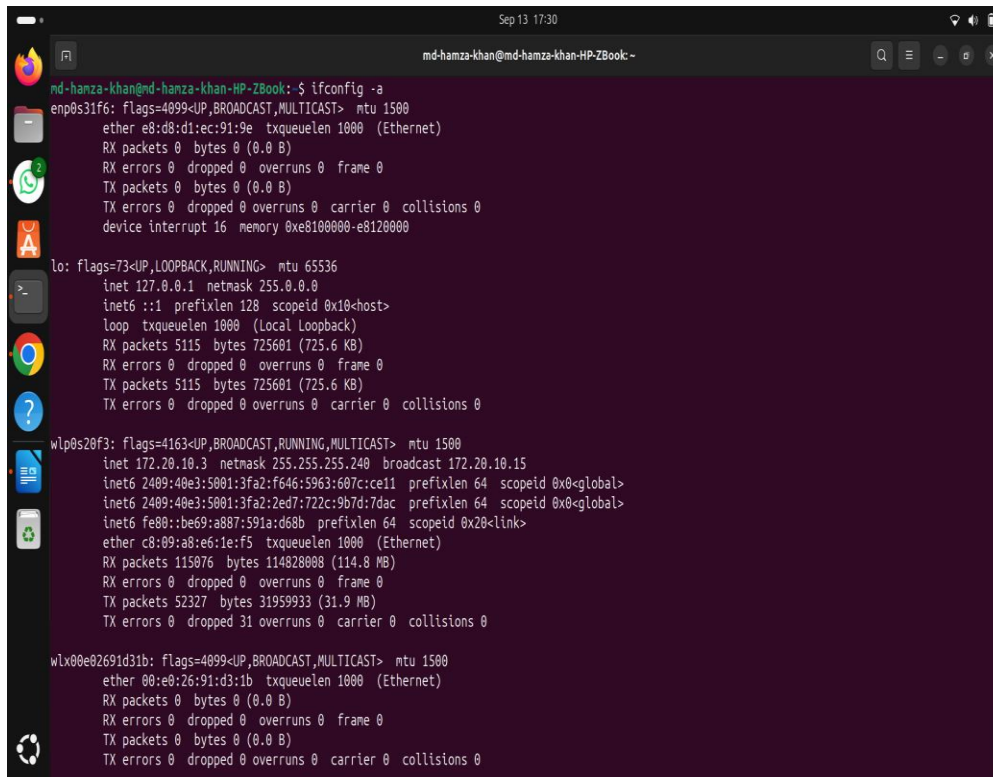
16. Loopback test result (Task 3)

Running ping -c 4 127.0.0.1 returned 4 packets transmitted, 4 received, 0% packet loss, with round-trip times around 0.059-0.060 ms. This test verifies the local loopback interface and confirms that the machine's own TCP/IP stack is functioning correctly, independent of any external network.

17. IPv4 vs MAC address (Task 4)

Address Type	Layer / Purpose
IPv4 address	Layer 3 — logical addressing and routing between networks
MAC address	Layer 2 — local, hardware-level interface addressing

SCREENSHOTS



```
md-hamza-khan@md-hamza-khan-HP-ZBook: ~  
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ifconfig -a  
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether e8:d8:d1:ec:91:9e txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 16 memory 0xe0100000-e8120000  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 5115 bytes 725601 (725.6 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 5115 bytes 725601 (725.6 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlp0s20f3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 172.20.10.3 netmask 255.255.255.240 broadcast 172.20.10.15  
    inet6 2409:40e3:5001:3fa2:f646:5963:607c:ce11 prefixlen 64 scopeid 0x0<global>  
    inet6 2409:40e3:5001:3fa2:2ed7:722c:9b7d:7dac prefixlen 64 scopeid 0x0<global>  
    inet6 fe80::be69:a887:591a:d68b prefixlen 64 scopeid 0x20<link>  
    ether c8:09:a8:e6:1e:f5 txqueuelen 1000 (Ethernet)  
    RX packets 115076 bytes 114820000 (114.8 MB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 52327 bytes 31959933 (31.9 MB)  
    TX errors 0 dropped 31 overruns 0 carrier 0 collisions 0  
  
wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether 00:e0:26:91:d3:1b txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 1 – `ifconfig -a` (IPv4 address, subnet mask, adapter)

```

md-hamza-khan@md-hamza-khan-HP-ZBook:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (enp0s31f6)
    Current Scopes: none
    Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported

Link 3 (wlp0s20f3)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 172.20.10.1
    DNS Servers: 172.20.10.1 fe80::e85f:2ff:fe2f:7664

Link 5 (wlx00e02691d31b)
    Current Scopes: none
    Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.059 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.060 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.059 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.059/0.059/0.060/0.000 ms
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ ping -c 4 google.com
PING google.com (2404:6800:4013:813::65) 56 data bytes
64 bytes from lcdels-in-f101.1e100.net (2404:6800:4013:813::65): icmp_seq=1 ttl=106 time=45.0 ms
64 bytes from lcdels-in-f101.1e100.net (2404:6800:4013:813::65): icmp_seq=2 ttl=106 time=40.3 ms
64 bytes from lcdels-in-f101.1e100.net (2404:6800:4013:813::65): icmp_seq=3 ttl=106 time=45.7 ms
64 bytes from lcdels-in-f101.1e100.net (2404:6800:4013:813::65): icmp_seq=4 ttl=106 time=51.2 ms

--- google.com ping statistics ---

```

Figure 2 – resolvectl status, ping 127.0.0.1, ping google.com

```
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 40.305/45.574/51.210/3.867 ms
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.29.138
Name:   google.com
Address: 142.250.29.102
Name:   google.com
Address: 142.250.29.101
Name:   google.com
Address: 142.250.29.113
Name:   google.com
Address: 142.250.29.139
Name:   google.com
Address: 142.250.29.100
Name:   google.com
Address: 2404:6800:4013:813::71
Name:   google.com
Address: 2404:6800:4013:813::65
Name:   google.com
Address: 2404:6800:4013:813::64
Name:   google.com
Address: 2404:6800:4013:813::8b

md-hamza-khan@md-hamza-khan-HP-ZBook:~$
```

Figure 3 – nslookup google.com

COMMANDS USED

```
ip a / ifconfig -a
resolvectl status
ping -c 4 127.0.0.1
ping -c 4 google.com
nslookup google.com
```

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Platform: Ubuntu Linux